

Unit 1: Introduction to GIS

1.1 GIS Introduction

Geographic Information System (GIS) is a system designed to capture, store, manipulate, analyze, manage, and present **spatial or geographic data**. It is a computer-based tool that helps people analyze, visualize, and interpret data related to locations on the Earth's surface. It combines **maps** with **data** to show patterns, relationships, and trends in the form of maps, charts, and reports. It integrates hardware, software, and data for processing geographic information.

Functions of GIS:

- Data collection and input
- Data storage and management
- Data analysis (e.g., overlay, buffer, spatial query)
- Data visualisation (maps, charts, reports)

Applications of GIS:

- Urban planning
- Environmental management
- Disaster risk assessment
- Transportation planning
- Agriculture and forestry

1.2 Scope of GIS

GIS is used in multiple fields for problem-solving and decision-making based on spatial data:

Sector	GIS Use Case
Environment	Monitoring pollution, deforestation
Health	Mapping disease outbreaks
Agriculture	Soil mapping, crop management(Soil preparation, Irrigation and water management, Harvesting and post-harvest handling)
Disaster Management	Risk zone mapping, evacuation planning
Business	Site selection, market analysis
Government	Cadastral(boundaries, ownership, and value of land and property) mapping, land use planning

GIS Scope Includes:

- **Real-time decision making:** Supports dynamic responses in emergencies, traffic management, and resource allocation
- **Integration with IoT and remote sensing technologies:** Enables automation, real-time monitoring, and large-scale spatial data analysis
- **Resource management:** Helps manage natural resources like water, forests, and minerals effectively
- **Land use planning and zoning:** Supports sustainable development by analyzing land suitability and urban growth patterns
- **Infrastructure and utility management:** Aids in managing networks like water supply, electricity, gas pipelines, and telecom
- **Environmental impact analysis:** Assesses changes in land use, pollution levels, and habitat loss
- **Public health tracking:** Maps disease outbreaks and healthcare access
- **Disaster preparedness and mitigation:** Identifies vulnerable areas and supports risk assessment
- **Transportation and logistics:** Optimizes routes, reduces travel time, and enhances delivery systems
- **Market analysis and site selection:** Assists businesses in identifying optimal locations and customer distribution
- **Climate change studies:** Monitors changes in climate variables and supports policy development

1.3 Think Spatially

Spatial thinking means understanding the **location, scale, shape, size, patterns, and relationships** between features or events on the Earth's surface—both **physical** (like rivers, roads, and buildings) and **abstract** (like population density or economic activity). It involves answering questions like:

- *Where is it located?*
- *What is around it?*
- *How are places connected?*

This way of thinking helps in making informed decisions, solving location-based problems, and interpreting maps or spatial data effectively.

Example:

Analyzing flood-prone areas by examining elevation data, rainfall patterns, and settlement locations.

Types of Spatial Data:**1. Vector Data**

- Represents geographic features using **points, lines, and polygons**.
- Each feature has precise coordinates.
- **Examples:**
 - **Points:** Locations of cities, wells, or landmarks
 - **Lines:** Roads, rivers, or utility networks
 - **Polygons:** Boundaries of countries, lakes, or land parcels
- **Advantages:** Accurate representation of discrete features; easy to analyze shapes and boundaries.

2. Raster Data

- Represents spatial data as a **grid of cells or pixels**, where each cell has a value.
- Often used for continuous data that varies over space.
- **Examples:**
 - Satellite images
 - Aerial photographs
 - Elevation or terrain maps
 - Land cover or temperature distribution
- **Advantages:** Ideal for representing continuous surfaces; good for image analysis and modeling.

Importance of Spatial Thinking:

- Helps solve location-based problems
- Supports better planning and resource management
- Enhances decision-making using maps and models